

CONCEPT 48.4

Neurons communicate with other cells at synapses

(pp. 1077–1083)

- In an electrical synapse, electrical current flows directly from one cell to another. In a chemical synapse, depolarization causes synaptic vesicles to fuse with the terminal membrane and release **neurotransmitter** into the synaptic cleft.
- At many synapses, the neurotransmitter binds to **ligand-gated ion channels** in the postsynaptic membrane, producing an **excitatory** or **inhibitory postsynaptic potential (EPSP or IPSP)**. The neurotransmitter then diffuses out of the cleft, is taken up by surrounding cells, or is degraded by enzymes. A single neuron has many synapses on its dendrites and cell body. Temporal and spatial **summation** of EPSPs and IPSPs at the axon hillock determine whether a neuron generates an action potential.
- Different receptors for the same neurotransmitter produce different effects. Some neurotransmitter receptors activate signal transduction pathways, which can produce long-lasting changes in postsynaptic cells. Major neurotransmitters include **acetylcholine**; the amino acids GABA, glutamate, and glycine; biogenic amines; **neuropeptides**; and gases such as NO.

❓ Why are many drugs that are used to treat nervous system diseases or to affect brain function targeted to specific receptors rather than particular neurotransmitters?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. What happens when a resting neuron's membrane depolarizes?
(A) There is a net diffusion of Na^+ out of the cell.
(B) The equilibrium potential for K^+ (E_K) becomes more positive.
(C) The neuron's membrane voltage becomes more positive.
(D) The cell's inside becomes more negative than the outside.
2. A common feature of action potentials is that they
(A) cause the membrane to hyperpolarize and then depolarize.
(B) can undergo temporal and spatial summation.
(C) are triggered by a depolarization that reaches threshold.
(D) move at the same speed along all axons.
3. Where are neurotransmitter receptors located?
(A) the nuclear membrane
(B) the nodes of Ranvier
(C) the postsynaptic membrane
(D) synaptic vesicle membranes

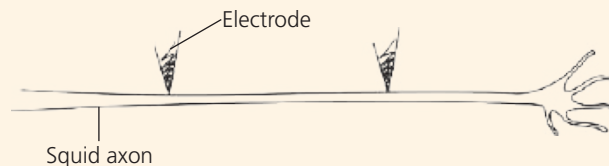
Levels 3-4: Applying/Analyzing

4. Why are action potentials usually conducted in one direction?
(A) Ions can flow along the axon in only one direction.
(B) The brief refractory period prevents reopening of voltage-gated Na^+ channels.
(C) The axon hillock has a higher membrane potential than the terminals of the axon.
(D) Voltage-gated channels for both Na^+ and K^+ open in only one direction.
5. Which of the following is the most *direct* result of depolarizing the presynaptic membrane of an axon terminal?
(A) Voltage-gated calcium channels in the membrane open.
(B) Synaptic vesicles fuse with the membrane.
(C) Ligand-gated channels open, allowing neurotransmitters to enter the synaptic cleft.
(D) An EPSP or IPSP is generated in the postsynaptic cell.

6. Suppose a particular neurotransmitter causes an IPSP in postsynaptic cell X and an EPSP in postsynaptic cell Y. A likely explanation is that
(A) the threshold value in the postsynaptic membrane for cell X is different from that for cell Y.
(B) the axon of cell X is myelinated, but that of cell Y is not.
(C) only cell Y produces an enzyme that terminates the activity of the neurotransmitter.
(D) cells X and Y express different receptor molecules for this particular neurotransmitter.

Levels 5-6: Evaluating/Creating

7. **WHAT IF?** Ouabain, a plant substance used in some cultures to poison hunting arrows, disables the sodium-potassium pump. What change in the resting potential would you expect to see if you treated a neuron with ouabain? Explain.
8. **WHAT IF?** If a drug mimicked the activity of GABA in the CNS, what general effect on behavior might you expect? Explain.
9. **DRAW IT** Suppose a researcher inserts electrodes at two positions along the middle of an axon dissected out of a squid. By applying a depolarizing stimulus, the researcher brings the plasma membrane at both positions to threshold. Using the drawing below as a model, create one or more drawings that illustrate where each action potential would terminate.



10. **EVOLUTION CONNECTION** An action potential is an all-or-none event. This on/off signaling is an evolutionary adaptation of animals that must sense and act in a complex environment. Imagine a nervous system in which the action potentials are graded, with the amplitude depending on the size of the stimulus. Describe what evolutionary advantage on/off signaling might have over this continuously variable kind of signaling.
11. **SCIENTIFIC INQUIRY** From what you know about action potentials and synapses, propose two hypotheses for how various anesthetics might block pain.
12. **WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), describe how the structure and electrical properties of vertebrate neurons reflect similarities and differences with other animal cells.
13. **SYNTHESIZE YOUR KNOWLEDGE**



This diamond-back rattlesnake (*Crotalus atrox*) alerts enemies to its presence with a rattle—a set of modified scales at the tip of its tail. Describe the roles of gated ion channels in initiating and moving a signal along the nerve from the snake's head to its tail and then to the muscle that shakes the rattle.

For selected answers, see Appendix A.